

PROJECT TITLE:

Egypt's Industrial Sector Performance Analysis

Project Timeline:

The project was completed over a period of **three months**, following a structured workflow to ensure high data quality and meaningful business insights.

Phase	Duration	Description
Business Understanding & Data Collection	2 Months	Defined the project objectives and key performance indicators (KPIs), researched and collected datasets from reliable sources, reviewed data quality, documented metadata, and prepared a comprehensive data dictionary to support the analysis.
Data Cleaning & Preparation	2 Weeks	Cleaned and transformed the datasets by handling missing values, removing duplicates, standardizing formats, correcting data types, integrating multiple data sources, and building a structured data model suitable for analysis.
Dashboard Development & Visualization	2 Weeks	Developed an interactive Power BI dashboard by creating DAX measures, designing visualizations, implementing filters and slicers, validating calculations, and optimizing the dashboard to effectively communicate insights on CO ₂ emissions, energy consumption, and economic performance.

Total Project Duration: 3 Months

Project Overview:

Problem Statement:

The core questions this project seeks to answer are:

Is Egypt's industrial growth occurring efficiently (decoupling economic growth from environmental impact), or is it happening at the expense of the environment — with emissions rising faster than economic growth?

Since Industry is one of the primary drivers of economic growth in Egypt, contributing significantly to GDP, employment, and exports. However, the industrial sector is simultaneously one of the largest consumers of energy and one of the biggest sources of carbon emissions in the economy. Understanding the relationship between industrial expansion, energy use, and emissions is essential for evaluating whether Egypt's current growth path is environmentally sustainable in the long term, and whether industrial expansion is being achieved through genuine efficiency gains or simply through increased resource consumption.

Key Analytical Questions:

This project is built around the following core questions:

- Is the growth rate of the industrial sector stable or volatile over time?
- Is the industrial sector's contribution to GDP increasing or decreasing?
- Do energy consumption and emissions grow at the same rate as industrial output, or slower?
- Is there measurable evidence of decoupling between industrial growth and CO₂ emissions?
- Does the industrial sector represent an environmental burden disproportionate to its economic weight?

Scope:

- **Time period covered:** 1990–2023
- **Geographic focus:** Egypt
- **Sectors included:** Industry, Electricity and heat production, Transport, Residential, Commercial and public services, Agriculture/Forestry, and other non-specified sectors.

Tools and Methodology:

The dashboard was built using **Power BI**, based on a data model constructed from multiple external sources, including the **World Bank**, the **International Energy Agency (IEA)**, and **CAPMAS** (used for cross-referencing and validation). All source tables are connected to a central calendar table, enabling consistent time-based filtering and year-over-year analysis across the entire model.

Deliverable

The final deliverable is an interactive four-page Power BI dashboard:

1. **Economic & Industrial Overview** — A comprehensive analysis of Egypt's economy and industrial performance.
2. **Economic & Industrial Performance** — How has Egypt's industrial sector contributed to economic performance over time?
3. **Energy Efficiency Analysis** — Is Egypt achieving industrial growth alongside higher energy efficiency?
4. **Environmental Sustainability** — Is Egypt achieving industrial growth while reducing its environmental impact?

Data Sources:

Primary Data Sources:

Three institutional sources were used to construct the dataset underlying this project:

1. World Bank (WB) :

Full name: The World Bank Open Data Repository

Role in this project: The World Bank served as the primary source for macroeconomic indicators relating to Egypt's overall economic output and the industrial sector's contribution to that output.

Data obtained from this source:

- Gross Domestic Product (GDP), expressed in constant 2015 US Dollars, to allow for inflation-adjusted comparison across the full time series.
- Industrial value added, expressed in constant 2015 US Dollars, representing the economic output generated specifically by the industrial sector.
- Industry's percentage contribution to total GDP, expressed as a decimal fraction (e.g., 0.3213 representing 32.13%).

2. International Energy Agency (IEA)

Full name: International Energy Agency

Role in this project: The IEA served as the primary source for all energy consumption and carbon emissions data used in this project, given its status as the internationally recognized authority on energy statistics.

Data obtained from this source:

- Total national CO₂ emissions from fuel combustion, expressed in million tons of CO₂ (MtCO₂).
- CO₂ emissions disaggregated by economic sector (Electricity and heat producers, Transport, Industry, Residential, Commercial and Public Services, Agriculture/Forestry, Other energy industries, and Non-specified/Other).
- Industrial energy consumption disaggregated by fuel source (Coal and coal products, Natural gas, Oil and oil products, Biofuels and waste, Electricity), expressed in terajoules (TJ).
- Total Final Consumption (TFC) of energy disaggregated by sector (Agriculture and forestry, Commercial and public services, Industry, Non-energy use, Other non-specified, Residential, Transport), expressed in terajoules (TJ).
- Sectoral share of Total Final Consumption (TFC), expressed as a decimal fraction of national energy demand.

3. Central Agency for Public Mobilization and Statistics (CAPMAS)

Full name: Central Agency for Public Mobilization and Statistics, Arab Republic of Egypt

Role in this project: CAPMAS was used as a supplementary and cross-referencing source, to validate the consistency and plausibility of the economic and industrial figures reported by the World Bank and IEA for the Egyptian context specifically.

Table-to-Source Mapping:

The following table provides a definitive mapping between each table in the Power BI data model and its corresponding source. This mapping was confirmed directly by the project author, with any previously unconfirmed sources resolved to the International Energy Agency (IEA), as instructed.

Table Name	Confirmed Source	Data Type
Total CO2 emissions - Egypt	IEA	Total national CO2 emissions from fuel combustion
CO2 emissions by sector - Egypt	IEA	CO2 emissions disaggregated by economic sector
EGY_GDP_Constant2015USD_WB	WORLD BANK	Gross Domestic Product, constant 2015 USD
EGY_Industry_Energy_BySource_1990_2024	IEA	Industrial energy consumption disaggregated by fuel source
EGY_Industry_PercentGDP_WB	WORLD BANK	Industry's percentage contribution to GDP
EGY_industry_ValueAdded_Constant2015USD_WB	WORLD BANK	Industrial value added, constant 2015 USD
Total final consumption (TFC) by sector - Egypt	IEA	Total final energy consumption disaggregated by sector
Share of total final consumption (TFC) by sector - Egypt	IEA	Sectoral share of total final energy consumption

Temporal Coverage

All data sources used in this project cover a consistent time period, ensuring comparability across the full dataset without the need for interpolation or period-adjustment:

- **Coverage period:** 1990–2023 (inclusive)
- **Granularity:** Annual (one observation per year per indicator)
- **Most recent available data point:** Calendar year 2023

Data cleaning and preparation :

Overview :

All data transformation and cleaning operations for this project were performed in Power Query in Power BI's Power Query Editor. Each source file — whether in CSV or Excel format — required a distinct sequence of cleaning steps due to structural differences in how the World Bank and IEA export their raw data (metadata header rows, wide year-by-year column layouts, inconsistent encodings). This section documents, table by table, the exact transformations applied, in the order they were applied, along with the rationale behind each step where evident from the transformation logic itself.

General Import Methodology

All source files were accessed via a folder connector (Folder.Files) pointing to a local directory, with each table's query filtering to its specific source file by name. This approach was used consistently across all IEA- and World Bank-derived CSV files.

A recurring structural issue across the raw CSV files is the presence of non-tabular metadata rows preceding the actual column headers (a standard characteristic of IEA and World Bank data exports, which typically include title lines, source attribution lines, and blank lines before the header row). Accordingly, nearly every query begins with a common pattern:

1. Import the raw CSV with all columns initially typed as text (to avoid type-conversion errors on metadata rows).
2. Skip the top rows containing non-tabular metadata (Table.Skip, typically 3 rows).
3. Promote the first remaining row to column headers (Table.PromoteHeaders).
4. Re-cast each column to its correct data type only after the true header row has been established.

This two-stage typing approach (text-first, then correctly-typed after header promotion) was applied specifically to prevent Power Query from throwing type-conversion errors against the metadata rows, which do not conform to the data types of the actual dataset.

CO2 Emissions by Sector — Egypt

Source file: CO2 emissions by sector - Egypt.csv (IEA)

Steps applied:

- Imported with 11 columns, Windows-1252 encoding.
- Skipped 3 metadata rows; promoted headers.
- Cast all sector columns to type number; the year column initially cast to Int64.Type.
- Removed the "Fishing" column — this sector was excluded from the analysis entirely (not retained in the model in any form).
- Reconstructed the year column as a proper date type by combining the extracted year integer with a fixed month/day of January 1st (#date([year],1,1)), replacing the original integer-year column with a true date column named year.
- Unpivoted all sector columns (Electricity and heat producers, Other energy industries, Transport Sector, Industry Sector, Commercial and Public Services, Residential, Agriculture/Forestry, Non-specified (Other)) into two columns: sector co2 (sector name) and Value co2 by sectore MtCO2 (emissions value), while explicitly preserving year and Units as fixed (non-unpivoted) columns.
- Removed the Units column after unpivoting (the unit, MtCO2, is documented in this report rather than retained as a data column).

Resulting structure: year, sector co2, Value co2 by sectore MtCO2 — a long-format (unpivoted) table used to support the Tree Map and Ribbon Chart visuals described in Section

The screenshot shows a data table with 26 rows and 4 columns. The columns are 'year', 'sector co2', '1.2 Value co2 by sectore MtCO2', and an unlabeled column. The data represents CO2 emissions by sector in Egypt from 1990 to 1993. The 'sector co2' column lists various sectors, and the 'Value co2 by sectore MtCO2' column shows the corresponding emissions values. The 'year' column shows dates in YYYY-MM-DD format.

	year	sector co2	1.2 Value co2 by sectore MtCO2
1	1/1/1990	Electricity and heat producers	22.381
2	1/1/1990	Other energy industries	2.981
3	1/1/1990	Transport Sector	16.149
4	1/1/1990	Industry Sector	26.538
5	1/1/1990	Commercial and Public Services	0.055
6	1/1/1990	Residential	9.743
7	1/1/1990	Agriculture/Forestry	0
8	1/1/1990	Non-specified (Other)	0.011
9	1/1/1991	Electricity and heat producers	23.824
10	1/1/1991	Other energy industries	3.082
11	1/1/1991	Transport Sector	16.901
12	1/1/1991	Industry Sector	24.125
13	1/1/1991	Commercial and Public Services	0.055
14	1/1/1991	Residential	9.263
15	1/1/1991	Agriculture/Forestry	0
16	1/1/1991	Non-specified (Other)	0.011
17	1/1/1992	Electricity and heat producers	24.544
18	1/1/1992	Other energy industries	3.089
19	1/1/1992	Transport Sector	17.566
20	1/1/1992	Industry Sector	25.732
21	1/1/1992	Commercial and Public Services	0.073
22	1/1/1992	Residential	9.083
23	1/1/1992	Agriculture/Forestry	0
24	1/1/1992	Non-specified (Other)	0.011
25	1/1/1993	Electricity and heat producers	24.359
26	1/1/1993	Other energy industries	3.102

The right side of the screenshot shows the 'Query Settings' panel. It includes a 'Name' field with the value 'CO2 emissions by sector - Egypt'. Below this is a list of 'APPLIED STEPS' with icons indicating the sequence of operations: Source, Navigation, Imported CSV, Changed Type, Removed Top Rows, Promoted Headers, Changed Type1, Removed Columns, Changed Type2, Renamed Columns, Changed Type3, Added Custom, Reordered Columns, Changed Type4, Removed Columns1, Renamed Columns1, Unpivoted Columns, Reordered Columns1, Renamed Columns2, and 'Removed Columns2' (which is highlighted with a red 'X' icon).

GDP (Constant 2015 USD) — World Bank

Source file: EGY_GDP_Constant2015USD_WB.csv (World Bank)

Steps applied:

- Imported with 70 columns, UTF-8 (65001) encoding — reflecting the World Bank's standard wide-format export (one column per year from 1960 onward, plus indicator metadata columns).
- Promoted headers, then skipped 3 additional metadata rows and re-promoted headers a second time (a two-pass header resolution, necessitated by an extra layer of title/attribution rows specific to this file's export format).
- Cast Country Name, Country Code, Indicator Name, Indicator Code to text, and all year columns (1960–2025) to number.
- Filtered rows to Country Name = "Egypt, Arab Rep." — this is the step that isolates Egypt-specific data from what is otherwise a global dataset covering all World Bank member countries.
- Unpivoted all year columns (1960–2025) into two columns: Attribute (year) and Value (GDP figure), while preserving Country Name.
- Skipped 3 further rows post-unpivot (removing residual non-data rows generated by the unpivot operation).
- Cast Attribute to date type and Value to Int64.Type; renamed Attribute to year.
- Skipped 29 additional rows, then skipped 1 more row — together, these two skip operations remove all years prior to the project's analysis window, retaining only data from 1990 onward (consistent with the project's confirmed scope of 1990–2023).
- Renamed Value to gdp value.
- Removed the last row of the table (Table.RemoveLastN, 1 row) — this removes a trailing row corresponding to a year beyond the dataset's actual coverage (e.g., a forecast/placeholder year with no real data).
- Removed the Country Name column (no longer needed, as the table is now Egypt-specific by construction).

Resulting structure: year, gdp value.

Query Settings

Table.RemoveColumns("#Removed Bottom Rows",{"Country Name"})

	year	gdp value
1	1/1/1990	1.15166E+11
2	1/1/1991	1.16462E+11
3	1/1/1992	1.21671E+11
4	1/1/1993	1.25201E+11
5	1/1/1994	1.30179E+11
6	1/1/1995	1.36219E+11
7	1/1/1996	1.43014E+11
8	1/1/1997	1.50869E+11
9	1/1/1998	1.59281E+11
10	1/1/1999	1.68923E+11
11	1/1/2000	1.79683E+11
12	1/1/2001	1.86035E+11
13	1/1/2002	1.90482E+11
14	1/1/2003	1.96565E+11
15	1/1/2004	2.04609E+11
16	1/1/2005	2.13758E+11
17	1/1/2006	2.28387E+11
18	1/1/2007	2.44575E+11
19	1/1/2008	2.62078E+11
20	1/1/2009	2.74326E+11
21	1/1/2010	2.88446E+11
22	1/1/2011	2.93536E+11
23	1/1/2012	3.00071E+11
24	1/1/2013	3.06629E+11
25	1/1/2014	3.1557E+11
26	1/1/2015	3.29367E+11

Query Settings

PROPERTIES

Name

EGY_GDP_Constant2015USD_WB

APPLIED STEPS

Source

Navigation

Imported CSV

Promoted Headers

Changed Type

Removed Top Rows

Promoted Headers1

Changed Type1

Filtered Rows

Unpivoted Other Columns

Removed Top Rows1

Reordered Columns

Changed Type2

Renamed Columns

Removed Top Rows2

Removed Top Rows3

Renamed Columns1

Removed Bottom Rows

Removed Columns

Industrial Energy Consumption by Source (1990–2024)

Source file: EGY_Industry_Energy_BySource_1990_2024.csv (IEA)

Steps applied:

- Imported with 7 columns, Windows-1252 encoding.
- Skipped 3 metadata rows; promoted headers.
- Cast the year column to date, each fuel-source column (Coal and coal products, Natural gas, Biofuels and waste, Electricity, Oil and oil products) to Int64.Type, and Units to text.
- Renamed the year column to year.
- Unpivoted all five fuel-source columns into Energy Source TJ (fuel type) and Energy Value (consumption figure), preserving year and Units.
- Removed the Units column.

Query Settings

Table.RemoveColumns("#Renamed Columns1",{"Units"})

	year	Energy Source TJ	Energy Value
1	1/1/1990	Coal and coal products	14546
2	1/1/1990	Natural gas	54269
3	1/1/1990	Biofuels and waste	352
4	1/1/1990	Electricity	60749
5	1/1/1990	Oil and oil products	271063
6	1/1/1991	Coal and coal products	17405
7	1/1/1991	Natural gas	53028
8	1/1/1991	Biofuels and waste	207
9	1/1/1991	Electricity	62964
10	1/1/1991	Oil and oil products	232037
11	1/1/1992	Coal and coal products	17727
12	1/1/1992	Natural gas	53871
13	1/1/1992	Biofuels and waste	207
14	1/1/1992	Electricity	65257
15	1/1/1992	Oil and oil products	251719
16	1/1/1993	Coal and coal products	21260
17	1/1/1993	Natural gas	65269
18	1/1/1993	Biofuels and waste	207
19	1/1/1993	Electricity	67633
20	1/1/1993	Oil and oil products	263801
21	1/1/1994	Coal and coal products	18452
22	1/1/1994	Natural gas	36303
23	1/1/1994	Biofuels and waste	264
24	1/1/1994	Electricity	70095
25	1/1/1994	Oil and oil products	196444
26	1/1/1995	Coal and coal products	15573

Query Settings

PROPERTIES

Name

EGY_Industry_Energy_BySource_1990_202

APPLIED STEPS

Source

Navigation

Imported CSV

Changed Type

Removed Top Rows

Promoted Headers

Changed Type1

Renamed Columns

Unpivoted Columns

Renamed Columns1

Removed Columns

Resulting structure: year, Energy Source TJ, Energy Value

Industry's Percentage Contribution to GDP — World Bank

Source file: EGY_Industry_PercentGDP_WB.csv (World Bank)

Steps applied:

- Imported with 70 columns, UTF-8 encoding.
- Cast metadata columns to text and year columns to number; promoted headers.
- Cast all columns to their final types (Country Name/Code, Indicator Name/Code as text; years 1960–2025 as number).
- Filtered rows to Country Name = "Egypt, Arab Rep.".
- Unpivoted all year columns into Attribute (year) and Value (percentage figure), preserving Country Name.
- Skipped 3 rows post-unpivot; reordered columns; renamed Attribute to year.
- Cast year to date type and Value to Percentage.Type.
- Divided the Value column by 100 (Table.TransformColumns, each _ / 100) — this step converts the World Bank's raw percentage figures (stored as whole numbers, e.g., 32.13) into decimal fractions (e.g., 0.3213), consistent with Power BI's native percentage formatting convention. This is the definitive source-level correction of the percentage-scaling issue identified and resolved during dashboard construction (see Section 4 — Data Model, Efficiency and Environmental measures).
- Skipped 30 additional rows (isolating the 1990-onward analysis window).
- Removed the last row (trailing non-data row).
- Removed the Country Name column.

Resulting structure: year, Value (decimal fraction, e.g., 0.3213 = 32.13%).

The screenshot displays a Power BI Desktop interface. The main area shows a table with two columns: 'year' and '% Value'. The table contains 26 rows of data, starting from 1/1/1990 and ending at 1/1/2015. The values range from 27.41% to 36.63%. The table's formula bar shows the query: `Table.RemoveColumns(#"Removed Bottom Rows",{"Country Name"})`. On the right, the 'Query Settings' pane is open, showing the 'Name' as 'EGY_Industry_PercentGDP_WB'. Under the 'APPLIED STEPS' section, the following steps are listed: Source, Navigation, Imported CSV, Changed Type, Promoted Headers, Changed Type1, Filtered Rows, Unpivoted Other Columns, Removed Top Rows, Reordered Columns, Renamed Columns, Changed Type2, Divided Column, Removed Top Rows1, Removed Bottom Rows, and Removed Columns.

	year	% Value
1	1/1/1990	27.41%
2	1/1/1991	32.13%
3	1/1/1992	31.41%
4	1/1/1993	31.15%
5	1/1/1994	30.50%
6	1/1/1995	30.25%
7	1/1/1996	29.52%
8	1/1/1997	29.00%
9	1/1/1998	28.65%
10	1/1/1999	28.40%
11	1/1/2000	30.75%
12	1/1/2001	30.90%
13	1/1/2002	32.58%
14	1/1/2003	33.37%
15	1/1/2004	34.67%
16	1/1/2005	34.15%
17	1/1/2006	36.15%
18	1/1/2007	35.07%
19	1/1/2008	36.21%
20	1/1/2009	35.82%
21	1/1/2010	35.78%
22	1/1/2011	35.95%
23	1/1/2012	39.25%
24	1/1/2013	39.89%
25	1/1/2014	39.89%
26	1/1/2015	36.63%

Industrial Value Added (Constant 2015 USD) — World Bank

Source file: EGY_industry_ValueAdded_Constant2015USD_WB.csv (World Bank)

Steps applied:

- Imported with 71 columns, UTF-8 encoding (one additional trailing column compared to the GDP file, subsequently discarded).
- Promoted headers, skipped 3 metadata rows, re-promoted headers.
- Cast all metadata and year columns to their appropriate types, including a trailing unnamed text column.
- Filtered rows to Country Name = "Egypt, Arab Rep.".
- Unpivoted year columns into Attribute (year) and Value (value-added figure), preserving Country Name.
- Skipped 3 rows post-unpivot; cast Attribute to date; reordered columns; renamed Attribute to year.
- Cast Value to Int64.Type.
- Skipped 29 rows (isolating 1990-onward data, consistent with the GDP table).
- Renamed Value to industry by Value add.
- Removed the last 2 rows (Table.RemoveLastN, 2 rows) — note this differs from the GDP table, which removed only 1 trailing row; this is flagged in Section 3.4 for confirmation of intent.
- Removed the Country Name column.

Resulting structure: year, industry by Value add.

The screenshot shows a data table with 26 rows and 3 columns. The first column is labeled 'year' and contains dates from 1/1/1990 to 1/1/2015. The second column is labeled 'industry by Value add' and contains numerical values. The third column is empty. To the right of the table is a 'Query Settings' panel. The 'PROPERTIES' section shows the name 'EGY_industry_ValueAdded_Constant2015'. The 'APPLIED STEPS' section lists various operations: Source, Navigation, Imported CSV, Promoted Headers, Changed Type, Removed Top Rows, Promoted Headers1, Changed Type1, Filtered Rows, Unpivoted Other Columns, Removed Top Rows1, Changed Type2, Reordered Columns, Renamed Columns, Changed Type3, Removed Top Rows2, Renamed Columns1, Removed Bottom Rows, and Removed Columns (which is highlighted with a red 'X').

	year	industry by Value add
1	1/1/1990	42659175257
2	1/1/1991	44963460020
3	1/1/1992	47347497510
4	1/1/1993	48059575633
5	1/1/1994	49129814955
6	1/1/1995	51808163124
7	1/1/1996	54405115916
8	1/1/1997	57076864456
9	1/1/1998	59703515565
10	1/1/1999	63725044018
11	1/1/2000	67984271600
12	1/1/2001	73016834312
13	1/1/2002	73228868605
14	1/1/2003	74496322452
15	1/1/2004	76522567496
16	1/1/2005	79121500743
17	1/1/2006	88017771190
18	1/1/2007	94138335947
19	1/1/2008	1.01097E+11
20	1/1/2009	1.06541E+11
21	1/1/2010	1.11684E+11
22	1/1/2011	1.12162E+11
23	1/1/2012	1.13504E+11
24	1/1/2013	1.14925E+11
25	1/1/2014	1.18023E+11
26	1/1/2015	1.20648E+11

Share of Total Final Consumption (TFC) by Sector — Egypt :

Source file: Share of total final consumption (TFC) by sector - Egypt.csv (IEA)

Steps applied:

- Imported with 6 columns, Windows-1252 encoding.
- Skipped 3 metadata rows; promoted headers.
- Cast the year column to date, and all four share columns (Transport, Industry, Residential, Commercial and public services) to Percentage.Type.
- Divided each of the four share columns by 100 individually, in four separate transformation steps — converting the source's raw percentage figures into decimal fractions, identical in purpose to the correction applied to the Industry % of GDP table (Section 3.3.4).
- Removed the Units column.
- Renamed the year column to year.

Resulting structure: year, Transport share in TFC, Industry share in TFC, Residential share in TFC, Commercial and public services share in TFC — all expressed as decimal fractions

	year	% Transport share in TFC	% Industry share in TFC	% Residential share in TFC	% Commercial and public ser
1	1/1/1990	23.01%	41.09%	24.32%	
2	1/1/1991	24.66%	38.45%	24.70%	
3	1/1/1992	24.62%	39.36%	23.83%	
4	1/1/1993	24.51%	40.59%	21.92%	
5	1/1/1994	28.23%	32.82%	25.31%	
6	1/1/1995	27.57%	33.91%	24.51%	
7	1/1/1996	27.46%	34.73%	23.49%	
8	1/1/1997	27.41%	34.38%	23.63%	
9	1/1/1998	29.36%	31.61%	24.46%	
10	1/1/1999	29.48%	32.61%	22.28%	
11	1/1/2000	30.31%	31.12%	23.42%	
12	1/1/2001	27.34%	38.10%	22.75%	
13	1/1/2002	26.20%	33.32%	22.14%	
14	1/1/2003	27.88%	31.76%	22.88%	
15	1/1/2004	28.68%	26.33%	24.03%	
16	1/1/2005	24.17%	32.69%	21.22%	
17	1/1/2006	24.59%	32.54%	20.69%	
18	1/1/2007	24.52%	34.02%	20.20%	
19	1/1/2008	25.62%	31.46%	21.02%	
20	1/1/2009	27.08%	27.41%	22.24%	
21	1/1/2010	28.12%	27.47%	22.25%	
22	1/1/2011	28.24%	27.73%	22.42%	
23	1/1/2012	30.38%	24.37%	22.50%	
24	1/1/2013	31.30%	26.43%	23.43%	
25	1/1/2014	33.31%	24.63%	24.66%	
26	1/1/2015	27.95%	24.07%	25.40%	

Query Settings

PROPERTIES

Name

Share of total final consumption (TFC) by

All Properties

APPLIED STEPS

Source

Navigation

Imported CSV

Changed Type

Removed Top Rows

Promoted Headers

Changed Type1

Divided Column

Divided Column1

Divided Column2

Divided Column3

Removed Columns

Renamed Columns

Total Final Consumption (TFC) by Sector — Egypt

Source file: Total final consumption (TFC) by sector - Egypt (1).csv (IEA)

This source file was processed through three separate Power Query queries, each applying a different final transformation to the same underlying wide-format data. All three share an identical initial sequence:

- Imported with 9 columns, Windows-1252 encoding.
- Skipped 3 metadata rows; promoted headers.
- Cast the year column to date, and each sector column (Industry, Transport, Residential, Commercial and public services, Agriculture and forestry, Other non-specified, Non-energy use) to Int64.Type.
- Renamed the year column to year.
- Duplicated the Industry column as Industry1, then renamed the original Industry column and the duplicate such that Industry1 retains the original industry values and a new Industry column exists as a second copy — effectively preserving an untouched copy of the Industry figures before the unpivot step below alters the table's shape.
- Reordered columns to position Industry1 immediately after year (excluded from the subsequent unpivot) while the duplicate Industry column remains among the columns eligible for unpivoting.

= Table.ReorderColumns("#Renamed Columns1",{ "year", "Industry1", "Transport", "Residential", "Commercial and public services", "Agriculture and forestry", "Other non-specified", "Non-energy use" })						
year	Industry1	Transport	Residential	Commercial and public services	Agriculture and forestry	Other non-specified
1/1/1990	400981	224572	237343	99	99	99
1/1/1991	365642	234493	234867	99	99	99
1/1/1992	388783	243165	235328	131	131	131
1/1/1993	418172	252530	225874	134	134	134
1/1/1994	321559	276626	247968	151	151	151
1/1/1995	360588	293152	260626	194	194	194
1/1/1996	392483	310305	265453	3474	3474	3474
1/1/1997	414044	330093	284609	4183	4183	4183
1/1/1998	394392	366294	305104	4483	4483	4483
1/1/1999	430698	389346	294346	4995	4995	4995
1/1/2000	411628	400844	309705	5351	5351	5351
1/1/2001	540568	387869	322777	5860	5860	5860
1/1/2002	508633	399909	337892	6406	6406	6406
1/1/2003	480996	422268	346549	6851	6851	6851
1/1/2004	397327	432886	362671	7529	7529	7529
1/1/2005	576915	426455	374470	8153	8153	8153
1/1/2006	614136	464024	390474	8773	8773	8773
1/1/2007	712955	514020	423340	9342	9342	9342
1/1/2008	668297	544362	446551	9682	9682	9682
1/1/2009	588829	581819	477894	10297	10297	10297
1/1/2010	609089	623501	493499	10496	10496	10496
1/1/2011	639904	651599	517477	10757	10757	10757
1/1/2012	570251	710938	526589	12627	12627	12627
1/1/2013	605276	716736	536371	14087	14087	14087
1/1/2014	552703	747463	553275	13747	13747	13747
1/1/2015	550516	767540	502265	13705	13705	13705

Industrial Energy by Source, with CO2 (Transformed — Excel Source)

Source file: EGY_Industry_Energy_BySource_1990_2024_transformed.xlsx, sheet "Egypt Industry Energy"

Steps applied:

- Imported directly from an Excel workbook (Excel.Workbook), navigating to the named sheet.
- Promoted headers.
- Cast columns: Year (Int64), Units (text), Energy Source (text), Energy Value (Int64), CO2 (Mt) (number).
- Removed the Units column.
- Renamed Energy Value to Energy Value tj.

Resulting structure: Year, Energy Source, Energy Value tj, CO2 (Mt).

= Table.RenameColumns(#"Removed Columns",{{"Energy Value", "Energy Value tj"}})				
12_3 Year	A8_C Energy Source	12_3 Energy Value tj	1_2 CO2 (Mt)	
1	1990 Coal and coal products	14546	1.3760516	
2	1990 Natural gas	54269	3.0444909	
3	1990 Biofuels and waste	352	0	
4	1990 Electricity	60749	null	
5	1990 Oil and oil products	271063	19.8689179	
6	1991 Coal and coal products	17405	1.646513	
7	1991 Natural gas	53028	2.9748708	
8	1991 Biofuels and waste	207	0	
9	1991 Electricity	62964	null	
10	1991 Oil and oil products	232037	17.0083121	
11	1992 Coal and coal products	17727	1.6769742	
12	1992 Natural gas	53871	3.0221631	
13	1992 Biofuels and waste	207	0	
14	1992 Electricity	65257	null	
15	1992 Oil and oil products	251719	18.4510027	
16	1993 Coal and coal products	21260	2.011196	
17	1993 Natural gas	65269	3.6615909	
18	1993 Biofuels and waste	207	0	
19	1993 Electricity	67633	null	
20	1993 Oil and oil products	263801	19.3366133	
21	1994 Coal and coal products	18452	1.7455592	
22	1994 Natural gas	36303	2.0365983	
23	1994 Biofuels and waste	264	0	
24	1994 Electricity	70095	null	
25	1994 Oil and oil products	196444	14.3993452	
26	1995 Coal and coal products	15573	1.4732058	

Query Settings

PROPERTIES

Name

Egypt Industry Energy

All Properties

APPLIED STEPS

Source

Navigation

Promoted headers

Changed column type

Removed Columns

Renamed Columns

Data Modeling:

Overview

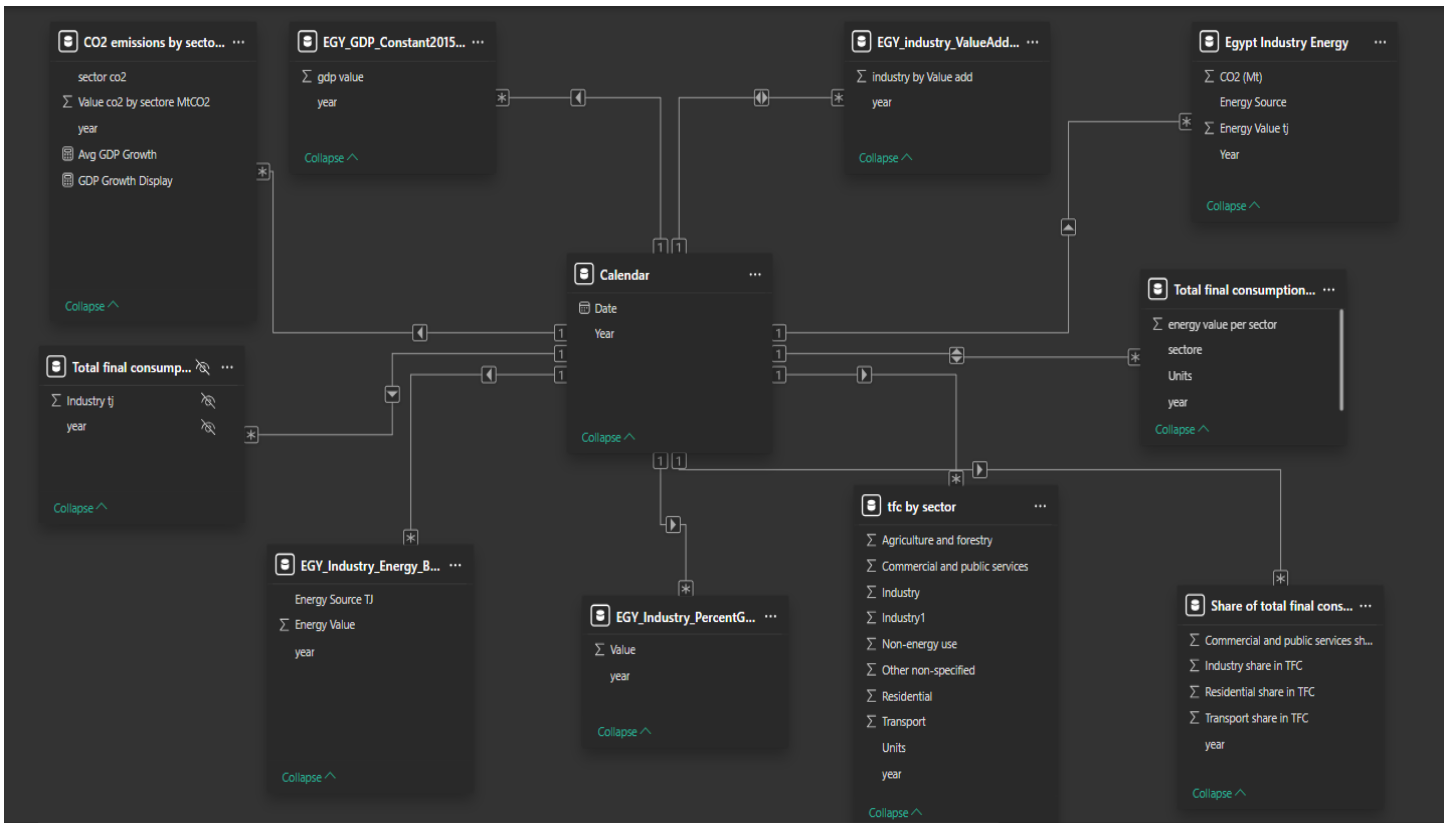
The Power BI data model consists of **10 tables**, connected through a central **Calendar** dimension table using the **Year** field as the common join key. This star-schema-like structure allows all fact tables — regardless of their original source or shape — to be filtered consistently by a single Year slicer across every page of the dashboard. All relationships in the model are **Active** (no inactive/dotted relationships exist), meaning every table is available for direct filtering without requiring **USERELATIONSHIP()** in any DAX measure.

Relationships

The following relationships connect the model's fact tables to the central Calendar dimension table, all via the Year field:

- **Calendar** has a **one-to-many** relationship with **CO2 emissions by sector - Egypt**.
- **Calendar** has a **one-to-many** relationship with **EGY_GDP_Constant2015USD_WB**.
- **Calendar** has a **one-to-many** relationship with **EGY_industry_ValueAdded_Constant2015USD_WB**.
- **Calendar** has a **one-to-many** relationship with **Egypt Industry Energy**.
- **Calendar** has a **one-to-many** relationship with **EGY_Industry_Energy_BySource**.
- **Calendar** has a **one-to-many** relationship with **EGY_Industry_PercentGDP_WB**.
- **Calendar** has a **one-to-many** relationship with **Total final consumption**.
- **Calendar** has a **one-to-many** relationship with **tfc by sector**.
- **Calendar** has a **one-to-many** relationship with **Share of total final consumption**.

All ten tables connect to the central **Calendar** table through the **Year** column, using active one-to-many relationships (one Calendar year to many fact rows). This design ensures that a single Year slicer, placed on any report page, filters every visual across the dashboard consistently, regardless of which underlying table each visual is built from.



Design Notes

- The **CO2 emissions by sector** - Egypt table hosts two GDP-related measures (**Avg GDP Growth**, **GDP Growth Display**) alongside its CO2 data, grouping related growth-analysis measures for convenient access during report building.
- The **Total final consumption...** table retains two hidden columns (**Industry tj**, **year**), preserved in the model as a lightweight, industry-only reference extracted from the broader TFC-by-sector dataset.
- The **tfc by sector** table is the fully unpivoted, long-format version of the TFC-by-sector dataset, retaining both the original **Industry** column and a duplicated **Industry1** column alongside all other sector categories, **Units**, and **year**.
- The **Egypt Industry Energy** table, sourced from Excel, extends the industrial energy dataset with a **CO2 (Mt)** column, providing fuel-source-level emissions data alongside energy consumption figures.

Table Inventory

- **Calendar** (Dimension)
Columns: **Date**, **Year**
- **CO2 emissions by sector - Egypt** (Fact — unpivoted)
Columns: **sector co2**, **Value co2 by sector MtCO2**, **year**
Measures: **Avg GDP Growth**, **GDP Growth Display**
- **EGY_GDP_Constant2015USD_WB** (Fact)
Columns: **gdp value**, **year**
- **EGY_industry_ValueAdded_Constant2015USD_WB** (Fact)
Columns: **industry by Value add**, **year**
- **Egypt Industry Energy** (Fact — Excel source)
Columns: **CO2 (Mt)**, **Energy Source**, **Energy Value tj**, **Year**
- **EGY_Industry_Energy_BySource** (Fact — unpivoted)
Columns: **Energy Source TJ**, **Energy Value**, **year**
- **EGY_Industry_PercentGDP_WB** (Fact)
Columns: **Value**, **year**
- **Total final consumption...** (Fact — hidden columns)
Columns: **Industry tj (hidden)**, **year (hidden)**
- **tfc by sector** (Fact — unpivoted)
Columns: **Agriculture and forestry**, **Commercial and public services**, **Industry**, **Industry1**, **Non-energy use**, **Other non-specified**, **Residential**, **Transport**, **Units**, **year**
- **Share of total final consumption...** (Fact)
Columns: **Commercial and public services share...**, **Industry share in TFC**, **Residential share in TFC**, **Transport share in TFC**, **year**

Measures (DAX) :

- **Totals**

These four measures form the foundational building blocks of the entire model. Every other measure in the model is ultimately derived from one or more of these four values, so their accuracy is critical to the validity of the entire analysis.

1. CO2 Industry

```
CO2 Industry =  
CALCULATE(  
    SUM('CO2 emissions by sector - Egypt'[Value co2 by sectore MtCO2]),  
    'CO2 emissions by sector - Egypt'[sector co2] = "Industry Sector"  
)
```

Unit: Million tonnes of CO2 (MtCO2)

Mechanical explanation:

The underlying `CO2 emissions by sector - Egypt` table is stored in long (unpivoted) format, meaning every sector's emissions value for every year occupies its own row, identified by the `sector co2` column (e.g., "Industry Sector", "Transport Sector", "Residential", etc.). This measure uses `CALCULATE` to apply a filter restricting the `SUM` to only those rows where `sector co2` equals exactly "Industry Sector", before summing the corresponding emissions values. In any year context (e.g., when filtered by the Calendar's Year slicer), this returns a single number: total industrial emissions for that year.

Purpose:

This measure exists specifically to isolate the industrial sector's carbon footprint from the seven other sectors present in the emissions dataset (Electricity and heat producers, Transport, Residential, Commercial and Public Services, Agriculture/Forestry, Other energy industries, Non-specified). Without this isolation, any emissions figure used in the analysis would represent the whole economy, diluting the relationship between industrial output and industrial pollution with the behavior of unrelated sectors. This measure is the emissions-side anchor for every environmental and decoupling calculation in the project — it answers the question "how much CO2 does industry itself produce?" as opposed to "how much CO2 does Egypt as a whole produce?"

2. GDP Value

```
GDP Value =  
SUM(EGY_GDP_Constant2015USD_WB[gdp value])
```

Unit: US Dollars, constant 2015 prices

Mechanical explanation:

A straightforward aggregation of the `gdp value` column from the World Bank GDP table. Because the source data is already inflation-adjusted (expressed in constant 2015 USD), no further currency conversion is required — the sum, filtered to a single year via the Calendar relationship, returns that year's national GDP.

Purpose:

While the current Decoupling measures (Section 5.4) no longer use GDP directly as their comparison benchmark, GDP Value remains in the model as essential macroeconomic context. It allows the dashboard to answer contextual questions such as "how large is the industrial sector relative to the whole economy?" (via Industry Value ÷ GDP Value) and provides the reader with a sense of the overall scale of Egypt's economy against which industrial figures can be mentally benchmarked, even where it is not used in a specific ratio.

3. Industry Energy

```
Industry Energy =  
SUM('Total final consumption (TFC) by sector - Egypt (1'[Industry tj])
```

Unit: Terajoules (TJ)

Mechanical explanation:

Sum, the `Industry tj` column from the Total Final Consumption (TFC) by sector table — a column that, per the ETL documentation, was deliberately preserved as an untouched, industry-only extraction of the wider TFC dataset before that dataset was unpivoted for other uses (see Section 3.3.7, Query Variant A). Each year's context, this returns the total energy consumed by the industrial sector across all fuel types combined.

Purpose:

This is the energy-side anchor of the analysis, representing how much energy input the industrial sector requires to operate. It is the denominator in every energy-efficiency ratio in the model (Energy Intensity, Growth Efficiency Ratio) and the basis for measuring whether industrial growth is being achieved through genuine efficiency gains or simply through consuming more energy.

4. Industry Value

```
Industry Value =  
SUM(EGY_industry_ValueAdded_Constant2015USD_WB[industry by Value add])
```

Unit: US Dollars, constant 2015 prices

Mechanical explanation:

Sums the `industry by Value add` column, which represents the economic value generated specifically by the industrial sector (as distinct from total GDP, which includes agriculture, services, and all other sectors).

Purpose:

This is the single most important measure in the entire model — it represents "industrial output" or "industrial growth" in economic terms, and serves as the numerator or comparison point in nearly every other measure in the model (Decoupling CO2, Decoupling Energy, Carbon Intensity, Energy Intensity, Growth Efficiency Ratio). Every question the project asks — is industry growing efficiently, is industry decoupling from pollution, is industry's environmental footprint proportionate to its economic contribution — ultimately traces back to this measure as the definition of "industrial growth" itself.

- **Growth**

These five measures convert the raw Totals into year-over-year percentage changes, which is the form in which growth, energy use, and emissions can be meaningfully compared to one another (since raw absolute values are on entirely different scales and cannot be compared directly).

CO2 Growth YOY %

```
CO2 Growth YOY % =  
DIVIDE(  
    [CO2 Industry] -  
    CALCULATE(  
        [CO2 Industry],  
        SAMEPERIODLASTYEAR('Calendar'[Date])  
    ),  
    CALCULATE(  
        [CO2 Industry],  
        SAMEPERIODLASTYEAR('Calendar'[Date])  
    )  
)
```

Unit: Percentage

Mechanical explanation:

SAMEPERIODLASTYEAR('Calendar'[Date]) shifts the current filter context back exactly one year, allowing CALCULATE to compute [CO2 Industry] as it existed in the prior year. The formula then applies the standard percentage-change calculation: (current year – prior year) ÷ prior year, wrapped in DIVIDE to safely return blank rather than an error if the prior year's value is zero or missing.

Purpose:

This measure answers "did industrial emissions rise or fall this year, and by how much?" It is the direct climate-impact indicator for the industrial sector and is a required input into the Decoupling CO2 measure (Section 5.4), where it is compared against the rate of industrial output growth.

Energy Growth YOY %

```
Energy Growth YOY % =  
DIVIDE(  
    [Industry Energy] -  
    CALCULATE(  
        [Industry Energy],  
        SAMEPERIODLASTYEAR('Calendar'[Date])  
    ),  
    CALCULATE(  
        [Industry Energy],  
        SAMEPERIODLASTYEAR('Calendar'[Date])  
    )  
)
```

Unit: Percentage

Mechanical explanation:

Identical logic to CO2 Growth YOY %, applied instead to [Industry Energy].

Purpose:

Answers "did industrial energy consumption rise or fall this year, and by how much?" This is the resource-consumption counterpart to CO2 Growth — while CO2 Growth measures pollution output, Energy Growth measures resource input. Comparing the two against each other (implicitly, since both derive from the same industrial activity) helps distinguish whether emissions changes are driven by changes in energy volume or by changes in the carbon intensity of the energy mix.

GDP Growth YOY %

```
GDP Growth YOY % =  
DIVIDE(  
    [GDP Value] -  
    CALCULATE([GDP Value], SAMEPERIODLASTYEAR('Calendar'[Date])),  
    CALCULATE([GDP Value], SAMEPERIODLASTYEAR('Calendar'[Date]))  
)
```

Unit: Percentage

Mechanical explanation:

Same year-over-year pattern applied to national GDP.

Purpose:

Provides the whole-economy growth rate as macroeconomic context, allowing the dashboard to show, for example, how industrial growth compares to overall national economic growth in any given year — useful for understanding whether industry is growing faster or slower than the economy as a whole, even though this comparison is no longer the basis of the Decoupling measures themselves.

Industry Growth YOY %

```
Industry Growth YOY % =  
DIVIDE(  
    [Industry Value] -  
    CALCULATE(  
        [Industry Value],  
        SAMEPERIODLASTYEAR(Calendar[Date])  
    ),  
    CALCULATE(  
        [Industry Value],  
        SAMEPERIODLASTYEAR(Calendar[Date])  
    )  
)
```

Unit: Percentage

Mechanical explanation:

Same year-over-year pattern applied to [Industry Value].

Purpose:

This is the central growth benchmark of the entire analysis — it represents how fast the industrial sector's own economic output is expanding or contracting, independent of the rest of the economy. It is the figure against which both CO2 Growth and Energy Growth are compared in the Decoupling measures, making it the single most-referenced growth measure in the model.

Growth Efficiency Ratio

```
Growth Efficiency Ratio =  
DIVIDE([Industry Growth YOY %], [Energy Growth YOY %])
```

Unit: Ratio (dimensionless)

Mechanical explanation:

Divides the industrial output growth rate by the industrial energy growth rate for the same year.

Purpose:

This measure provides a single, easily interpretable number summarizing the relationship between output growth and energy growth, as an alternative framing to the subtraction-based Decoupling Energy measure. A ratio greater than 1 means industrial value added is growing faster than industrial energy consumption (i.e., each additional unit of energy is producing more economic output than before — an efficiency improvement). A ratio between 0 and 1 means energy consumption is growing faster than output (efficiency is declining). A negative ratio indicates the two measures are moving in opposite directions (e.g., output growing while energy consumption falls, or vice versa), which requires case-by-case interpretation of the underlying Growth measures to understand correctly. This measure is particularly useful for dashboard visuals such as gauges or KPI cards where a single "efficiency score" is more immediately readable than a percentage-point difference.

- **Efficiency**

These two measures are the project's primary decoupling indicators — the direct, quantitative answer to the project's central research question.

Decoupling CO2

```
Decoupling CO2 =  
[Industry Growth YOY %] - [CO2 Growth YOY %]
```

Unit: Percentage points (difference of two percentages)

Mechanical explanation:

Subtracts the industrial CO2 growth rate from the industrial output growth rate for the same year.

Purpose:

This is the single most important measure in the entire project. It directly tests whether industrial growth is being achieved sustainably:

- A **positive** value means industrial output is growing faster than industrial emissions — industry is producing more economic value per unit of pollution than before, which is the definition of decoupling.
- A value of **zero** means output and emissions are growing at the same rate — no decoupling is occurring, but the situation is not worsening relative to output.
- A **negative** value means industrial emissions are growing faster than industrial output — industry is becoming more carbon-intensive per unit of economic value generated, the least sustainable scenario.

Unlike earlier versions of this measure (which compared national GDP growth to national CO2 growth), this version compares industrial growth to industrial CO2 growth specifically, making it a sector-internal test: is industry, on its own terms, becoming cleaner or dirtier as it grows?

Decoupling Energy

```
Decoupling Energy =  
[Industry Growth YOY %] - [Energy Growth YOY %]
```

Unit: Percentage points (difference of two percentages)

Mechanical explanation:

Subtracts the industrial energy growth rate from the industrial output growth rate for the same year.

Purpose:

This is the resource-efficiency counterpart to Decoupling CO₂. A positive value indicates industry is growing its output faster than its energy consumption — evidence of genuine improvement of genuine efficiency (doing more with proportionally less energy). A negative value indicates energy consumption is outpacing output growth, suggesting industrial expansion is being driven by higher resource consumption rather than improved productivity. Because energy consumption is the direct driver of most industrial emissions, Decoupling Energy often serves as a leading indicator: an improving trend here frequently precedes an improving trend in Decoupling CO₂, especially where the industrial fuel mix remains relatively stable year to year.

- **Environmental**

These three measures normalize industrial emissions and energy use against economic output, producing "intensity" indicators that describe efficiency at a point in time (as opposed to the Growth and Efficiency measures, which describe change over time).

Carbon Emission Factor

```
Carbon Emission Factor =  
DIVIDE(  
    [CO2 Industry] * 1000000,  
    [Industry Energy]  
)
```

Unit: tons of CO₂ per terajoule (tCO₂/TJ)

Mechanical explanation:

[CO₂ Industry] is stored in million tons (MtCO₂), so it is multiplied by 1,000,000 to convert to metric tons before being divided by [Industry Energy], which is already expressed in TJ. The result expresses how many tons of CO₂ are released for every TJ of energy consumed by industry.

Purpose:

This measure isolates the "cleanliness" of the industrial energy mix itself, independent of how much energy is being used or how much output is being produced. If this value falls over time, it indicates that industry is shifting toward lower-carbon fuel sources (e.g., more natural gas or renewables, less coal or heavy fuel oil) — a structural improvement in the emissions profile of energy use, as opposed to simply using less energy overall. If this value rises even while total emissions or energy use is falling, it will indicate the fuel mix is becoming dirtier even as scale shrinks, which is an important nuance the Growth and Decoupling measures alone cannot reveal.

Carbon Intensity (tCO₂ / Million USD)

Carbon Intensity tCO₂ / Million USD =

```
DIVIDE(  
  [CO2 Industry] * 1000000000,  
  [Industry Value]  
)
```

Unit: Tones of CO₂ per million US Dollars of industrial value added

Mechanical explanation:

[CO₂ Industry] (in MtCO₂) is multiplied by 1,000,000,000 — this larger multiplier (compared to Carbon Emission Factor) accounts for the fact that [Industry Value] is stored in whole USD rather than millions of USD, so the conversion simultaneously converts MtCO₂ to tones (×1,000,000) and USD to million-USD (an additional ×1,000 in the numerator to compensate for not dividing the denominator down to millions). The result expresses tons of CO₂ emitted per million dollars of industrial economic output produced.

Purpose:

This is the definitive "how clean is industrial growth" indicator and is one of the most widely used metrics internationally for assessing the environmental efficiency of economic activity (analogous to national-level Carbon Intensity of GDP but applied specifically to the industrial sector). A falling trend over the 1990–2023 period would represent the clearest possible evidence that Egypt's industrial sector is decoupling economic value creation from carbon emissions — producing more economic value for every ton of CO₂ released.

Energy Intensity (toe / Million USD)

```
Energy Intensity (toe / Million USD) =  
DIVIDE(  
    [Industry Energy] * 1000000,  
    [Industry Value]  
)
```

Unit: tons of oil equivalent per million US Dollars of industrial value added

Mechanical explanation:

[Industry Energy] (in TJ) is multiplied by 1,000,000 to align its scale with [Industry Value] (in whole USD), producing a result expressed as energy consumed per million dollars of industrial output.

Purpose:

This measure answers "how much energy does it take to generate a unit of industrial economic value?" the resource-efficiency counterpart to Carbon Intensity. A falling trend indicates industry is becoming more energy-efficient over time (extracting more economic value from each unit of energy consumed), which is a prerequisite for — though not a guarantee of — falling Carbon Intensity, since Carbon Intensity also depends on the cleanliness of the energy mix (captured separately by the Carbon Emission Factor measure above). Taken together, these three Environmental measures allow the analysis to distinguish between two distinct pathways to lower emissions: using less energy per unit of output (Energy Intensity) versus using cleaner energy per unit consumed (Carbon Emission Factor) with Carbon Intensity capturing their combined effect.

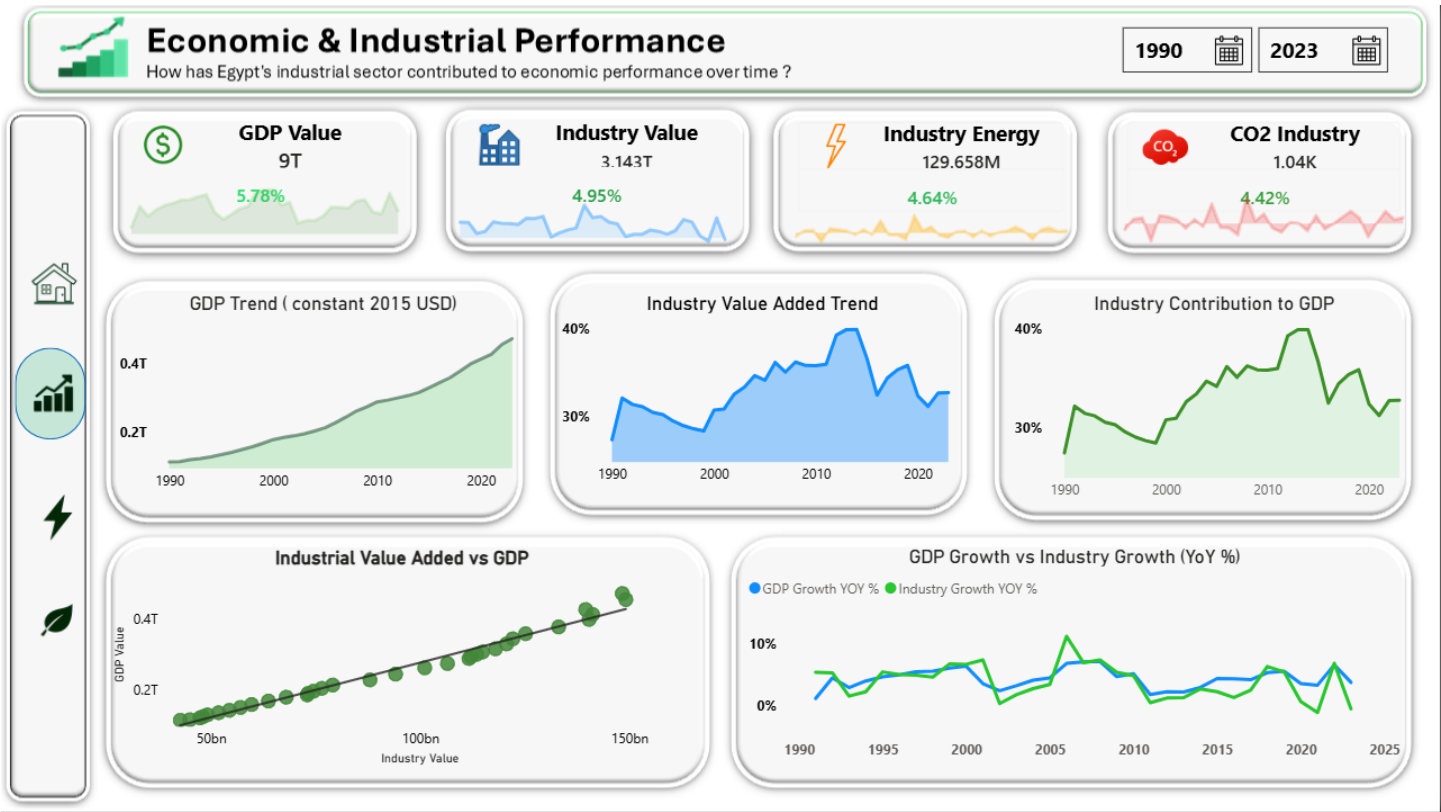
DASHBOARD & VISUALS:

Page 1 — Economic & Industrial Performance

Page title: Economic & Industrial Performance
Subtitle: "How has Egypt's industrial sector contributed to economic performance over time?"

Filter applied: Date range slicer, set to 1990–2023, positioned top-right of the page.

This page establishes the macro-to-micro economic narrative of the dashboard: it opens with headline totals, then shows how each core metric has trended over time, and closes with two relationship-focused visuals that test whether industrial growth and GDP growth move together.



KPI Cards (Top Row)



Four identical-structure KPI cards, each showing: an icon, a metric label, the current value, and a YoY growth percentage displayed beneath a small sparkline-style trend area.

Card	Measure (Value)	Measure (Growth %)	Icon
GDP Value	[GDP Value]	[GDP Growth YOY %]	Dollar sign
Industry Value	[Industry Value]	[Industry Growth YOY %]	Factory
Industry Energy	[Industry Energy]	[Energy Growth YOY %]	Lightning bolt
CO2 Industry	[CO2 Industry]	[CO2 Growth YOY %]	CO2 cloud

Steps to build (per card):

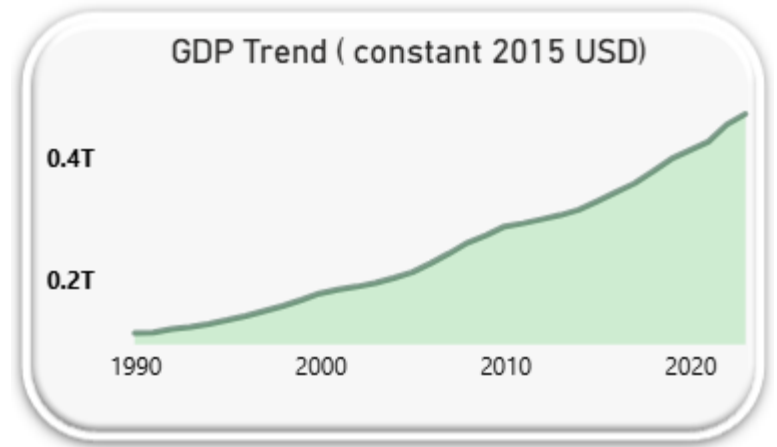
1. Insert a rounded-rectangle background shape for the card container.
2. Insert a small icon image in the top-left of the card (dollar sign, factory, lightning bolt, or CO2 cloud, matching the metric).
3. Add a **Card** visual (new-style Card, which supports a callout value plus a secondary label) and bind the **Fields** value to the relevant Totals measure (e.g., [GDP Value]).
4. Below the main card, overlay an **Area chart** (sparkline-style) with **Legend, Axis titles, and gridlines all turned off**, bound to `Calendar[Year]` on the axis and the corresponding measure as the value — sized small and positioned to sit inside the bottom portion of the card as a background trend visual.
5. Add a second small **Card** visual for the YoY growth percentage (e.g., [GDP Growth YOY %]), positioned over the sparkline, formatted in green text.
6. Apply **conditional formatting** (font color) to the growth-percentage card: green for positive values, red for negative — consistent with the color convention used throughout the dashboard.

GDP Trend (Constant 2015 USD)

Description: A green-shaded area chart showing the trajectory of national GDP from 1990 to roughly 2023, illustrating steady, accelerating growth.

Steps to build:

1. Insert an **Area chart** visual.
2. Axis: `Calendar[Year]`.
3. Values: `[GDP Value]`.
4. Format pane → Data colors: set the line/fill to green, with a light green semi-transparent fill beneath the line.
5. Format pane → Y-axis: enable **Display units = Auto** so the axis shows values like "0.2T", "0.4T" rather than full digit counts.
6. Title set to "GDP Trend (constant 2015 USD)" via Format pane → Title.

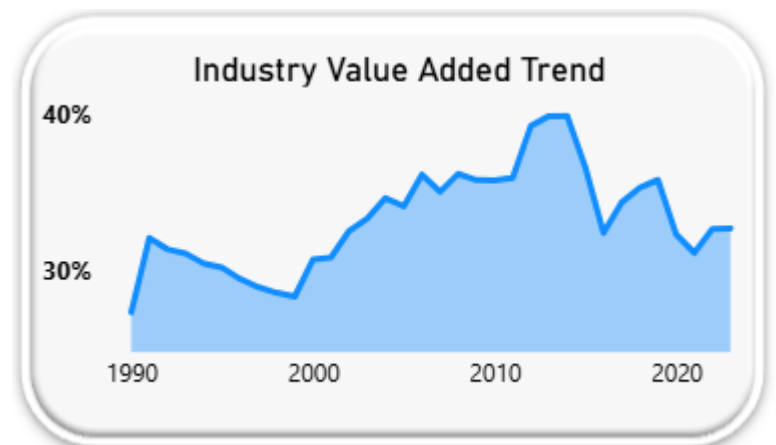


Industry Value Added Trend

Description: A blue-shaded area chart showing Industry's percentage contribution trend (labelled by axis values 30%–40%) over the same period, with visible year-to-year volatility layered on top of a rising trend.

Steps to build:

1. Insert an **Area chart** visual.
2. Axis: `Calendar[Year]`.
3. Values: bind to the measure representing industry's share/contribution trend (e.g., `[Industry % of GDP]` from the Totals measure table, formatted as Percentage).
4. Format pane → Data colors: set to blue with a light blue fill.
5. Title set to "Industry Value Added Trend".



Industry Contribution to GDP

Description: A green-line area chart, visually similar in shape to the previous chart but in a darker green tone, again plotted on a 30%–40% axis range — representing industry's contribution to GDP over time with a rising overall trend and cyclical dips.

Steps to build:

1. Insert an **Area chart** visual (or Line chart with fill enabled).
2. Axis: `Calendar[Year]`.
3. Values: the same or a closely related **industry-share-of-GDP** measure, distinguished from the previous chart by color and possibly a different level of smoothing.
4. Format pane → Data colors: set to a darker green tone to visually distinguish it from the blue chart beside it.
5. Title set to "Industry Contribution to GDP".

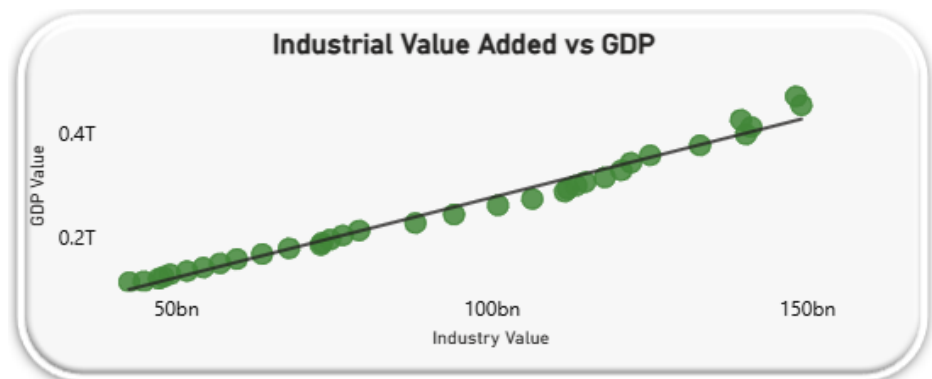


Industrial Value Added vs GDP (Scatter Plot)

Description: A scatter plot with Industry Value on the X-axis and GDP Value on the Y-axis, showing a strong, near-linear positive relationship (a fitted trend line rises consistently from bottom-left to top-right), with points colored dark green.

Steps to build:

1. Insert a **Scatter chart** visual.
2. X-axis: `[Industry Value]`.
3. Y-axis: `[GDP Value]`.
4. Values (used to force one data point per year rather than one aggregated point): add `Calendar[Year]` to the **Values** or **Details** field well, so each year renders as a separate dot.
5. Format pane → Data colors: set marker color to dark green.
6. Format pane → Trend line: enable **Analytics** → **Trend line** to render the fitted linear regression line shown across the scatter.
7. Title set to "Industrial Value Added vs GDP".

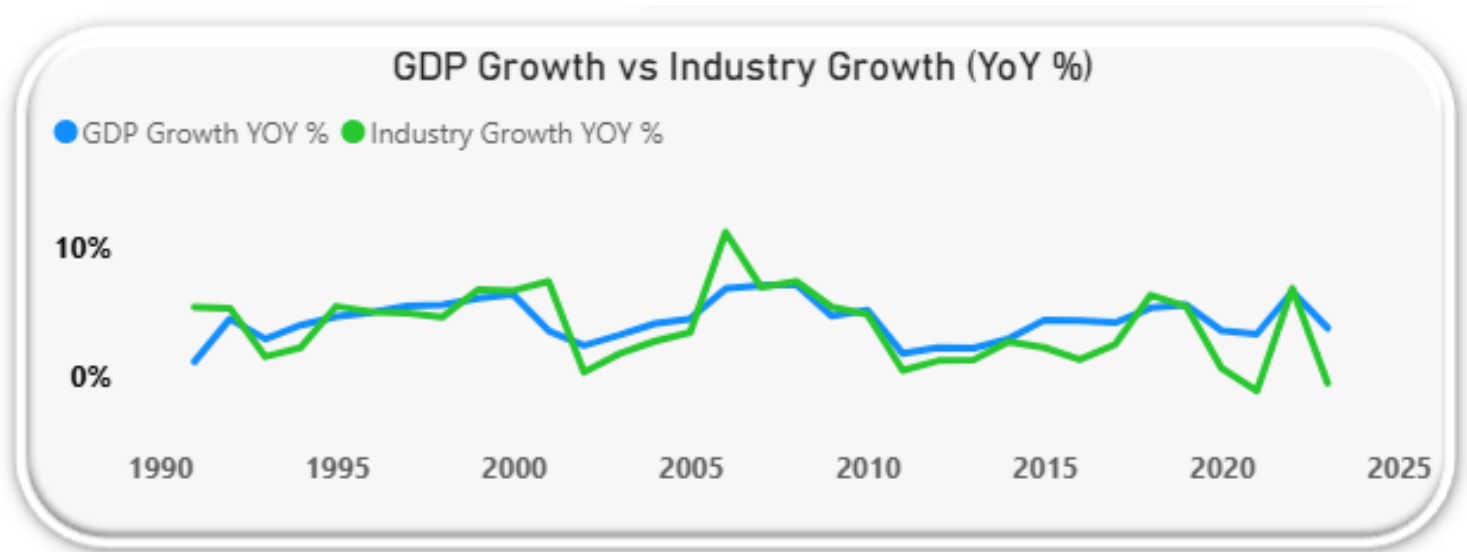


GDP Growth vs Industry Growth (YoY %)

Description: A dual-line chart comparing GDP Growth YOY % (blue) against Industry Growth YOY % (green) from 1990 to 2025, with both lines oscillating closely around and between 0%–10%, frequently crossing one another — visually illustrating that industrial growth tracks GDP growth closely but with greater short-term volatility.

Steps to build:

1. Insert a **Line chart** visual.
2. Axis: `Calendar[Year]`.
3. Values: add both `[GDP Growth YOY %]` and `[Industry Growth YOY %]`.
4. Format pane → Data colors: set GDP Growth YOY % to blue and Industry Growth YOY % to green, consistent with the color-coding used elsewhere on the page.
5. Format pane → Legend: position at top, matching the small colored dot + label style shown ("● GDP Growth YOY % ● Industry Growth YOY %").
6. Format pane → Y-axis: format as Percentage.
7. Title set to "GDP Growth vs Industry Growth (YoY %)".

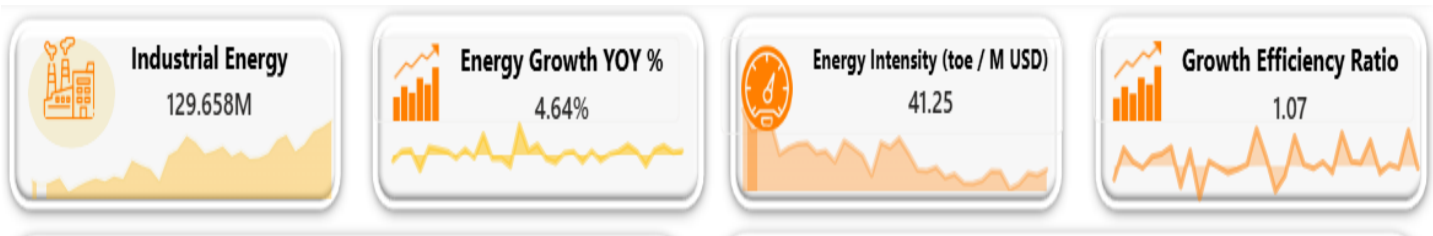


Page 2 — Energy Efficiency Analysis

Page title: Energy Efficiency Analysis
Subtitle: "Is Egypt Achieving Industrial Growth with Higher Energy Efficiency?"
Filter applied: Date range slicer, set to 1990–2023, positioned top-right of the page.

KPI Cards (Top Row)

Four identical-structure KPI cards, each showing: an icon, a metric label, the current value, and a background sparkline-style trend area utilizing an amber/orange color theme.



Card	Measure (Value)	Icon
Industrial Energy	[Industrial Energy]	Factory
Energy Growth YOY %	[Energy Growth YOY %]	Rising chart
Energy Intensity (toe / M USD)	[Energy Intensity]	Speedometer / Gauge
Growth Efficiency Ratio	[Growth Efficiency Ratio]	Rising chart

Steps to build (per card):

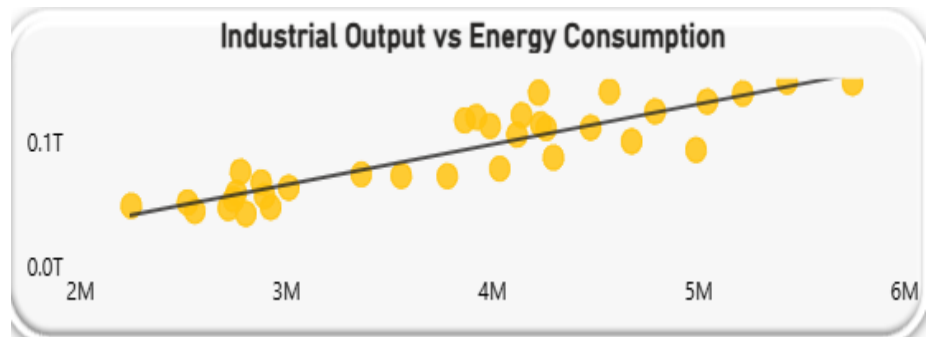
1. **Insert a rounded-rectangle background shape** for the card container with a subtle gray border and soft shadow.
2. **Insert a small icon image** in the top-left of the card (factory, rising chart, or gauge) matching the respective energy metric.
3. **Add a Card visual** (new-style Card) and bind the Fields value to the relevant measure (e.g., [Industrial Energy]). Format text to a bold, dark gray/black color.
4. **Overlay an Area chart (sparkline-style)** beneath the main value text. Turn off the Legend, Axis titles, labels, and gridlines. Bind Calendar[Year] to the X-axis and the corresponding metric to the Y-axis.
5. Set the area chart data color to **orange/amber with a semi-transparent gradient fill**, sized to sit cleanly within the bottom half of the card.

Industrial Output vs Energy Consumption (Scatter Plot)

Description: A scatter plot displaying the correlation between economic industrial output and total industrial energy consumption. Yellow data points represent individual years, tracing a strong, near-linear positive relationship from left to right along a fitted linear regression line.

Steps to build:

1. **Insert a Scatter chart** visual.
2. **X-axis:** Bind your industrial economic engine measure (e.g., [Industry Value]).
3. **Y-axis:** Bind the corresponding energy consumption measure (e.g., [Industrial Energy]).
4. **Values:** Add Calendar[Year] to the details field well to plot an individual marker for every historical year.
5. **Format pane → Markers:** Set the shape color to solid yellow/orange.
6. **Format pane → Analytics:** Expand the **Trend line** card and toggle it **On** to draw the linear regression line across the dataset.
7. **Title:** Set to "Industrial Output vs Energy Consumption".

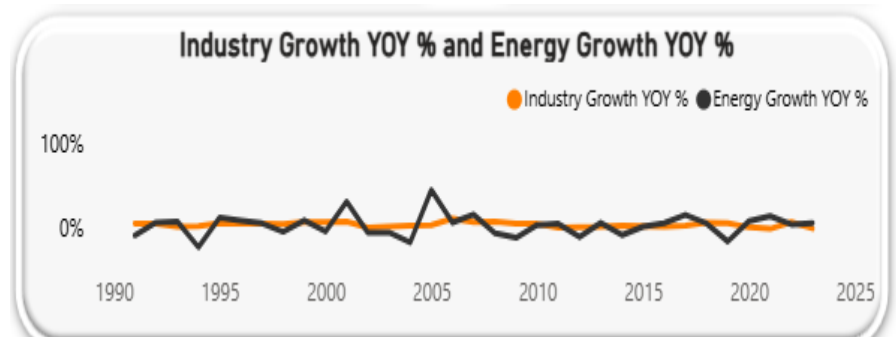


Industry Growth YOY % and Energy Growth YOY %

Description: A dual-line chart tracking and comparing the annual percentage changes of Industry Growth (orange line) against Energy Growth (black line) from 1990 to roughly 2023, exposing tracking volatility and decoupling gaps.

Steps to build:

1. **Insert a Line chart** visual.
2. **X-axis:** Calendar[Year].
3. **Y-axis:** Add both [Industry Growth YOY %] and [Energy Growth YOY %].
4. **Format pane → Lines:** Set [Industry Growth YOY %] to orange and [Energy Growth YOY %] to dark charcoal/black.
5. **Format pane → Legend:** Enable and position at the top-right using the simple text and marker style.
6. **Title:** Set to "Industry Growth YOY % and Energy Growth YOY %".

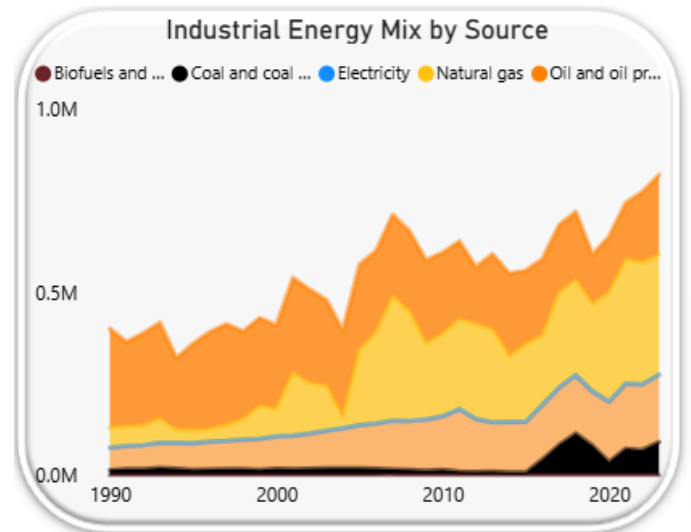


Industrial Energy Mix by Source

Description: A stacked area chart highlighting the historical volume and transition of industrial energy consumption broken down by source (Biofuels, Coal, Electricity, Natural Gas, Oil products) from 1990 to the present.

Steps to build:

1. **Insert a Stacked Area chart** visual.
2. **X-axis:** Calendar[Year].
3. **Legend:** Egypt Industry Energy[Energy Source].
4. **Y-axis:** [Energy Value].
5. **Format pane** → **Colors:** Assign custom thematic colors to the categories (e.g., deep orange for oil, bright yellow for natural gas, blue for electricity, and black for coal) to preserve a intuitive visual breakout.
6. **Title:** Set to "Industrial Energy Mix by Source".

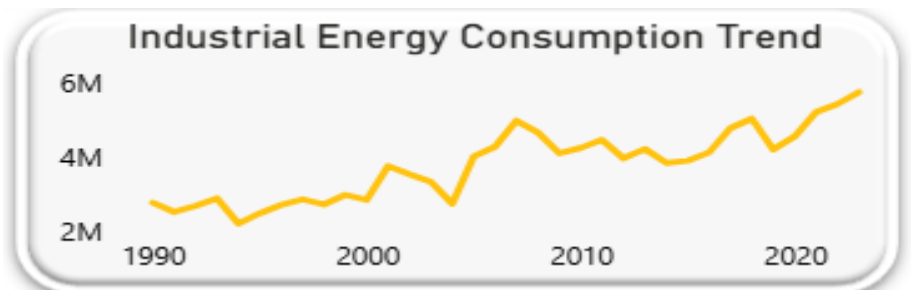


Industrial Energy Consumption Trend

Description: A clean, single-line trend chart tracking the raw upward trajectory of total industrial energy consumption in Egypt from 1990 to 2023.

Steps to build:

1. **Insert a Line chart** visual.
2. **X-axis:** Calendar[Year].
3. **Y-axis:** [Industrial Energy].
4. **Format pane** → **Lines:** Set color to a solid, vibrant yellow. Turn off gridlines for a cleaner look.
5. **Title:** Set to "Industrial Energy Consumption Trend".

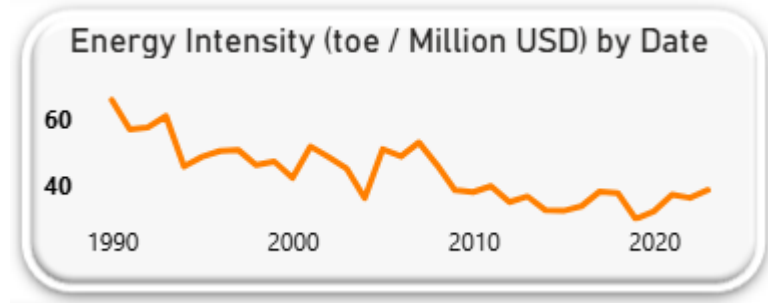


Energy Intensity (toe / Million USD) by Date

Description: A line chart displaying the long-term structural downward trend of energy intensity over time, visually demonstrating efficiency improvements despite overall industrial scaling.

Steps to build:

1. **Insert a Line chart** visual.
2. **X-axis:** `Calendar[Year]`.
3. **Y-axis:** `[Energy Intensity]`.
4. **Format pane** → **Lines:** Set line color to solid orange.
5. **Title:** Set to "Energy Intensity (toe / Million USD) by Date".

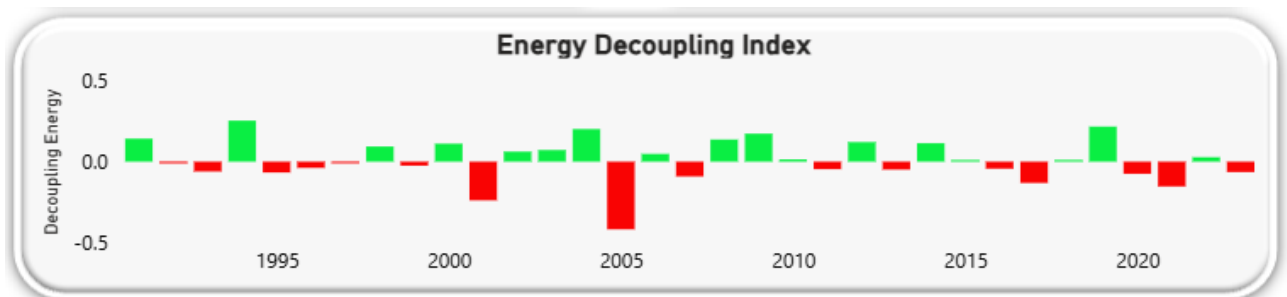


Energy Decoupling Index

Description: A positive/negative column chart plotting the yearly decoupling indicator. Positive values (green bars) highlight periods of successful efficiency decoupling, while negative values (red bars) spotlight years of resource inefficiency.

Steps to build:

1. **Insert a Clustered Column chart** visual.
2. **X-axis:** `Calendar[Year]`.
3. **Y-axis:** `[Decoupling Index]` (from your Efficiency/Environmental measure tables).
4. **Format pane** → **Columns** → **Conditional Formatting (fx):** * Set Format Style to **Rules**.
 - Base it on the field `[Decoupling Index]`.
 - **Rule 1:** If value is greater than or equal to 0 (Min) and less than 0, set color to **Red**.
 - **Rule 2:** If value is greater than or equal to 0 and less than or equal to Max, set color to **Green**.
5. **Title:** Set to "Energy Decoupling Index".



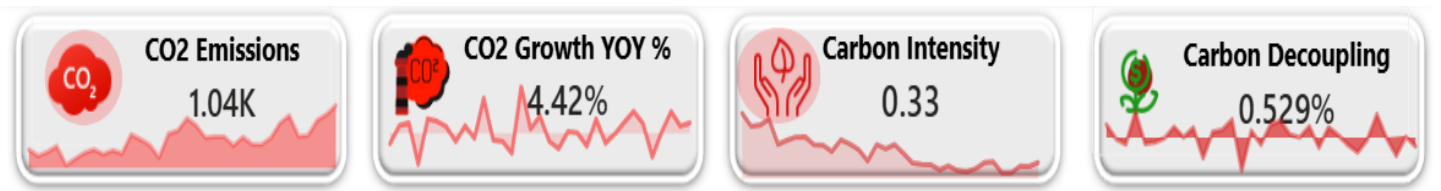
Page 3 — Environmental Sustainability

Page title: Environmental Sustainability
Subtitle: "Is Egypt Achieving Industrial Growth while reducing Environmental Impact?"
Filter applied: Date range slicer, set to 1990–2023, positioned top-right of the page.

KPI Cards (Top Row)

Four identical-structure KPI cards, each showing: an icon, a metric label, the current value, and a background sparkline-style trend area utilizing a soft red/crimson color theme.

Card	Measure (Value)	Icon
CO2 Emissions	[CO2 Industry] or [Value co2 by sectore]	CO2 Cloud
CO2 Growth YOY %	[CO2 Growth YOY %]	CO2 Cloud with Factory / Upward Trend
Carbon Intensity	[Carbon Intensity tCO2 / Million USD]	Hands cradling a leaf/sprout
Carbon Decoupling	[Decoupling CO2]	Sprout growing with currency/shield



Steps to build (per card):

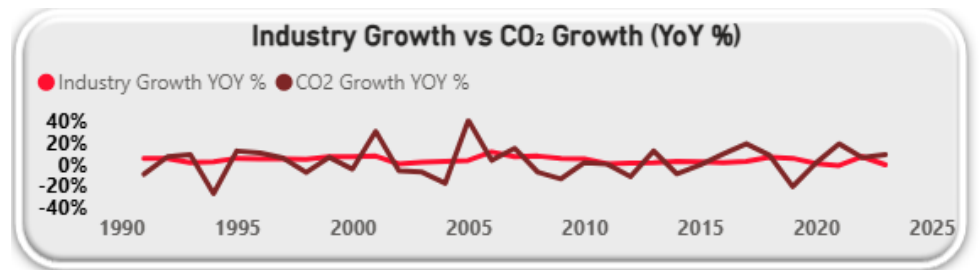
1. **Insert a rounded-rectangle background shape** for the card container with a subtle gray border and soft shadow.
2. **Insert a small icon image** in the top-left of the card (CO2 cloud, eco-hands, or environmental sprout) matching the respective metric.
3. **Add a Card visual** (new-style Card) and bind the Fields value to the relevant environmental measure. Format text to a bold, dark gray/black color.
4. **Overlay an Area chart (sparkline-style)** beneath the main value text. Turn off the Legend, Axis titles, labels, and gridlines. Bind `Calendar[Year]` to the X-axis and the corresponding metric to the Y-axis.
5. Set the area chart data color to **red/crimson with a semi-transparent gradient fill**, sized to sit cleanly within the bottom half of the card container

Industry Growth vs CO2 Growth (YoY %)

Description: A dual-line chart comparing Industry Growth YOY % (light red/brown line) against CO2 Growth YOY % (dark red line) from 1990 to 2025, illustrating the synchronization or variance between industrial acceleration and emissions.

Steps to build:

1. **Insert a Line chart** visual.
2. **X-axis:** Calendar[Year].
3. **Y-axis:** Add both [Industry Growth YOY %] and [CO2 Growth YOY %].
4. **Format pane** → **Lines:** Set Industry Growth YOY % to a lighter red-brown tone and CO2 Growth YOY % to dark crimson red.
5. **Format pane** → **Legend:** Position at the top-left using the small colored dot + label style.
6. **Format pane** → **Y-axis:** Enable gridlines, format as Percentage, and ensure bounds comfortably display ranges from -40% to 40%.
7. **Title:** Set to "Industry Growth vs CO2 Growth (YoY %)".

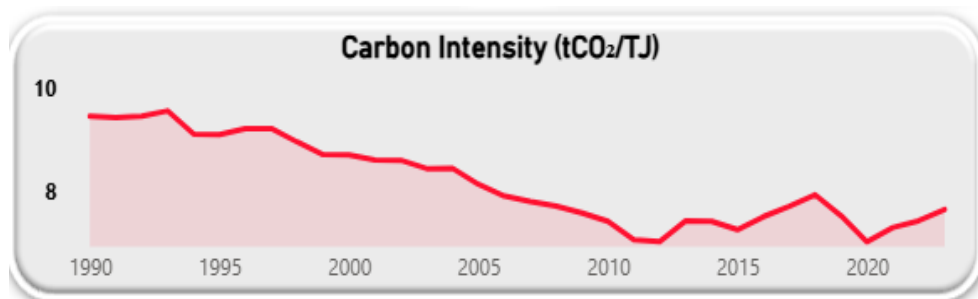


Carbon Intensity (tCO2/TJ)

Description: A line chart displaying the long-term trend of industrial carbon intensity over time, tracking whether industrial energy usage is becoming progressively cleaner.

Steps to build:

1. **Insert a Line chart** visual.
2. **X-axis:** Calendar[Year].
3. **Y-axis:** [Carbon Intensity tCO2 / Million USD] (or your target intensity measure).
4. **Format pane** → **Lines:** Set the line color to solid red.
5. **Format pane** → **Y-axis:** Set custom scale bounds (e.g., 6 to 10) if necessary to emphasize variations over the timeline.
6. **Title:** Set to "Carbon Intensity (tCO2/TJ)".

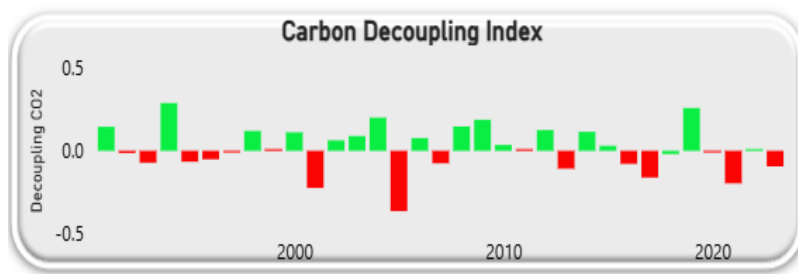


Carbon Decoupling Index

Description: A positive/negative clustered column chart plotting the yearly carbon decoupling performance indicator. Positive values (green bars) highlight periods of successful green growth, while negative values (red bars) isolate years where emissions grew faster than economic output.

Steps to build:

1. **Insert a Clustered Column chart** visual.
2. **X-axis:** **Calendar[Year]**.
3. **Y-axis:** **[Decoupling CO2]** (from your Efficiency/Environmental measure tables).
4. **Format pane** → **Columns** → **Conditional Formatting (fx)**:
 - Set Format Style to **Rules**.
 - Base it on the field [Decoupling CO2].
 - **Rule 1:** If value is greater than or equal to Min and less than 0, set color to **Red**.
 - **Rule 2:** If value is greater than or equal to 0 and less than or equal to Max, set color to **Green**.
5. **Title:** Set to "**Carbon Decoupling Index**".

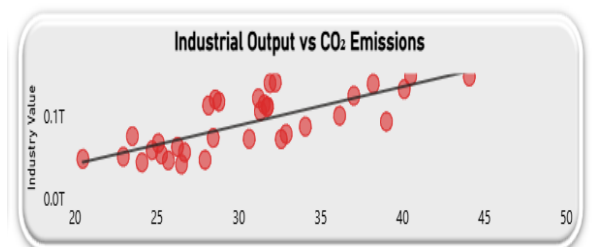


Industrial Output vs CO2 Emissions (Scatter Plot)

Description: A scatter plot illustrating the relationship between total Industry Value (Y-axis) and CO2 Emissions (X-axis). Individual years are plotted as semi-translucent red bubbles tracking along a positive linear regression trend line.

Steps to build:

1. **Insert a Scatter chart** visual.
2. **X-axis:** Bind your emissions measure (e.g., **[Value co2 by sectore]**).
3. **Y-axis:** Bind your industry economic engine measure (e.g., **[Industry Value]**).
4. **Values:** Drop **Calendar[Year]** into the details/values field well to generate one data point per year.
5. **Format pane** → **Markers:** Set the shape color to red and apply a slight transparency (~30%) so overlapping years remain legible.
6. **Format pane** → **Analytics:** Turn **Trend line** to **On** to plot the black linear regression tracking line.
7. **Title:** Set to "**Industrial Output vs CO2 Emissions**".

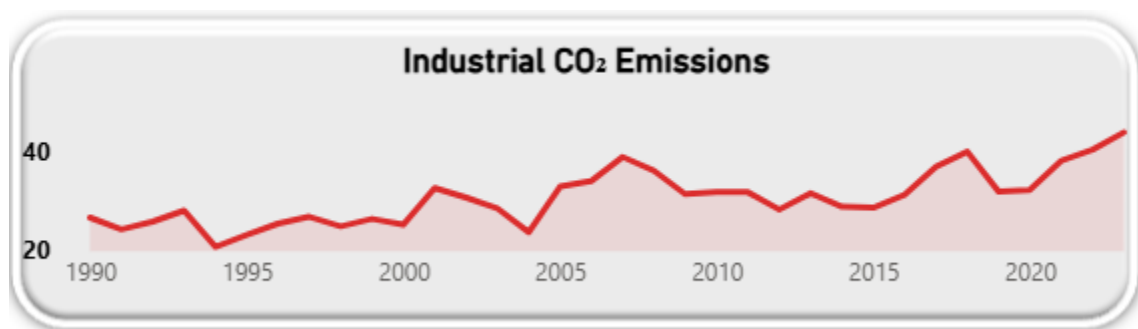


Industrial CO2 Emissions

Description: An area chart showcasing the direct, raw upward trajectory of industrial carbon dioxide emissions in Egypt from 1990 to roughly 2023.

Steps to build:

1. **Insert an Area chart** visual.
2. **X-axis:** Calendar[Year].
3. **Y-axis:** [Value co2 by sectore].
4. **Format pane** → **Data colors:** Set the line to solid red with a light red semi-transparent fill beneath it.
5. **Title:** Set to "Industrial CO2 Emissions".

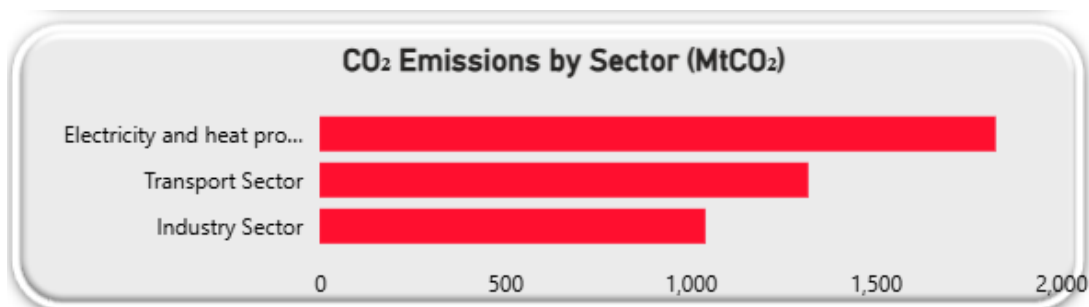


CO2 Emissions by Sector (MtCO2)

Description: A horizontal clustered bar chart stacking the total accumulated carbon emissions across Egypt's major tracking sectors (Electricity and heat production, Transport Sector, Industry Sector).

Steps to build:

1. **Insert a Clustered Bar chart** visual (horizontal alignment).
2. **Y-axis:** CO2 emissions by sector [sector co2].
3. **X-axis:** [Value co2 by sectore MtCO2].
4. **Format pane** → **Bars:** Set data color to a solid, vibrant red across all categories.
5. **Title:** Set to "CO2 Emissions by Sector (MtCO2)".



THANK YOU